

%APPENDIX

```
%% %Part1(a)
clc
clearvars

iggData= readtable("igg.xlsx");
age= iggData.age;
igg= iggData.IgG;
agesqr= age.^2;
iggData.agesqr= agesqr;

%Calculating the data for the summary
meanM1= [mean(age), mean(igg)];
medianM1= [median(age), median(igg)];
stdM1= [std(age), std(igg)];
maxM1= [max(age), max(igg)];
minM1= [min(age), min(igg)];

rowlabels= ['Age'; 'IgG'];

fprintf('                Summary                \n')
fprintf('_____ \n')
fprintf('      mean      median      std      max      min \n')
fprintf('_____ \n')
for i=1:length(meanM1)
    fprintf('%-6s %6.3f %6.3f %6.3f %6.3f %6.3f\n',...
        rowlabels(i,:), meanM1(i), medianM1(i), stdM1(i), maxM1(i), minM1(i))
end

skewAge= skewness(age)
skewIgg= skewness(igg)
figure
histogram(age)
xlabel('Age')
ylabel('Frequency')

figure
histogram(igg)
xlabel('IgG')
ylabel('Frequency')

%% %Part 1(b)
%Estimating model 1 using OLS
mdl1= fitlm(iggData, 'IgG~ age')

%% %Part 1(c)
%Estimating model 2 using OLS
X2= [age agesqr];
mdl2= fitlm(iggData, 'IgG~ age+ agesqr')

%% %Part 1(d)
x= [ones(size(age)), age, agesqr];
[b2, b2int, r2, r2int, stats2]=regress(igg, x);
beta_curr= [b2(1); b2(2); b2(3)];
maxiter=5;
tolerance= 1e-9;

my_hess= -(x' * x);

for i=1:maxiter
    fprintf('--- Iteration %d ---\n', i);
    fprintf('Current Beta: [%6.4f, %6.4f, %6.4f]\n', beta_curr);
    % gradient calculation
    gradient = x' * (igg - x*beta_curr);
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    %calculate the update step = -inv(H) * g(beta)
    step = - my_hess \ gradient;
    beta_new = beta_curr + step;
    change = norm(beta_new - beta_curr);
    if change < tolerance
        beta_curr = beta_new;
        fprintf("Convergance achieved");
        break;
    end
    beta_curr = beta_new;
end

resids= igg- x*beta_curr;

n= size(x,1);
k= size(x,2);

est_sigma= (resids'*resids)/(n-k);

cov_beta= est_sigma*inv(x'*x);

se= sqrt(diag(cov_beta));

res= table(beta_curr, se, 'RowNames', {'Intercept', 'Age', 'Agesqr'});
res.Properties.VariableNames= {'ML Estimates', 'SE'};
disp(res);

%% %Part 2(a)
clc
clearvars

data= readtable("marijuana.xlsx");
data= convertvars(data,
["q85", "educ", "income", "maritalStatus", "sex", "parent", "pastUse", "race", "party"], 'categorical');
data.y= zeros(size(data.q85));
data.y(data.q85=="Yes, legal")=1;
data.y(data.q85=="No, illegal")=0;

tabulate(data.y)
%% %Part 2(b)

data= renamevars(data, ["age", "hh1"], ["x2", "x3"]);

ageStats=[mean(data.x2) std(data.x2)]
hh1Stats=[mean(data.x3) std(data.x3)]
x2=data.x2;
x3=data.x3;

%% %Part 2(c)
data.x4= zeros(size(data.pastUse));
data.x4(data.pastUse=="Yes")=1;
data.x4(data.pastUse=="No")=0;
x4=data.x4;
tabulate(data.pastUse)
fprintf('\n')

%% %Part 2(d)

data.x5= zeros(size(data.sex));
data.x5(data.sex=="Male")=1;
data.x5(data.sex=="Female")=0;
tabulate(data.sex)
fprintf('\n')

data.x6= zeros(size(data.parent));

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data.x6(data.parent=="Yes")=1;
data.x6(data.parent=="No")=0;
tabulate(data.parent)
fprintf('\n')
x5=data.x5;
x6=data.x6;
%% %Part 2(e)

data.maritalStatus= renamecats(data.maritalStatus, "Never been married", "single");
data.maritalStatus= mergecats(data.maritalStatus, ["Divorced", "Separated", "Widowed"], "post-married");
data.maritalStatus= mergecats(data.maritalStatus, ["Married", "Living with a partner"], "couple");

data.x7= zeros(size(data.maritalStatus));
data.x7(data.maritalStatus=="single")=1;
data.x7(data.maritalStatus~= "single")=0;

data.x8= zeros(size(data.maritalStatus));
data.x8(data.maritalStatus=="post-married")=1;
data.x8(data.maritalStatus~= "post-married")=0;

x7=data.x7;
x8=data.x8;
tabulate(data.maritalStatus)
fprintf('\n')

%% %Part 2(f)

data.income= mergecats(data.income, ["Less than 10000", "10 to under 20000", ...
    "20 to under 30000", "30 to under 40000", "40 to under 50000"], "poor");
data.income= mergecats(data.income, ["50 to under 75000", "75 to under 100000"], "middle");
data.income= mergecats(data.income, ["100 to under 150000", "150000 or more"], "rich");

data.x9= zeros(size(data.income));
data.x9(data.income=="poor")=1;
data.x9(data.income~= "poor")=0;
x9=data.x9;

data.x10= zeros(size(data.income));
data.x10(data.income=="middle")=1;
data.x10(data.income~= "middle")=0;
x10=data.x10;
tabulate(data.income)
fprintf('\n')

%% %Part 2(g)

data.educ= mergecats(data.educ, ["Less than HS", "HS", "HS Incomplete"], "HSandBelow");
data.educ= mergecats(data.educ, ["Some college", "Associate Degree"], "lessThanBachelors");
data.educ= mergecats(data.educ, ["Bachelors", "Postgraduate Degree", "Some
Postgraduate"], "BachelorsandAbove");

data.x11= zeros(size(data.educ));
data.x11(data.educ=="HSandBelow")=1;
data.x11(data.educ~= "HSandBelow")=0;

data.x12= zeros(size(data.educ));
data.x12(data.educ=="lessThanBachelors")=1;
data.x12(data.educ~= "lessThanBachelors")=0;

tabulate(data.educ)
fprintf('\n')
x11=data.x11;
x12=data.x12;
%% %Part 2(h)

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data.race= renamecats(data.race,["White","Black"],["white", "black"]);
data.race= mergecats(data.race, ["Asian","Hispanic","Native American","Other Race","Pacific
Islander"],...
    "allOther");
data.x13= zeros(size(data.race));
data.x13(data.race=="white")=1;
data.x13(data.race~="white")=0;

data.x14= zeros(size(data.race));
data.x14(data.race=="black")=1;
data.x14(data.race~="black")=0;

tabulate(data.race)
fprintf('\n')
x13=data.x13;
x14=data.x14;
%% %Part 2(i)
data.party= renamecats(data.party, ["Democrat","Republican"],...
    ["democrat","republican"]);
data.party= mergecats(data.party,["(VOL) No preference","(VOL) Other party","Independent"],...
    "independentOthers");
data.x15 = zeros(size(data.party));
data.x15(data.party == "democrat") = 1;
data.x15(data.party ~= "democrat") = 0;

data.x16=zeros(size(data.party));
data.x16(data.party=="republican")=1;
data.x16(data.party~="republican")=0;

tabulate(data.party)
fprintf('\n')
x15=data.x15;
x16=data.x16;
%% %Part 3(b)
n= size(data,1);
y= data.y;

mdl= fitglm(data,'y ~ x2+x3+x4+x5+x6+x7+x8+x9+x10+x11+x12+x13+x14+x15+x16')
beta_estimates= mdl.Coefficients.Estimate;

%% Part 3(c)
X = [ones(size(y)), x2, x3, x4, x5, x6, x7, x8, x9, x10, x11, x12, x13, x14, x15, x16];

p_current= normcdf(X* beta_estimates);
X_new=X;
X_new(:,2)= X_new(:,2)+5;
p_new= normcdf(X_new*beta_estimates);
delta_p= mean(p_new- p_current);

fprintf('\n Change in probability (Age +5 years): %.4f\n', delta_p);

%% %Part 3(d)
X_1 = X;
X_0 = X;
X_1(:,6) = 1;
X_0(:,6) = 0;

p_1 = normcdf(X_1*beta_estimates);
p_0 = normcdf(X_0*beta_estimates);

delta_parents = mean(p_1 - p_0);
fprintf('\n Change in Probability if one is a parent is: %.4f\n', delta_parents);

%% %Part 4(b)

```

```
X_logit= [x2, x3, x4, x5, x6, x7, x8, x9, x10, x11, x12, x13, x14, x15, x16];
mdl_logit= fitglm(X_logit, y, 'Distribution','binomial','Link','logit')
```

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%% %Part 4(c)
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```
p_age = predict(mdl_logit, X_logit);
```

```
X_fitglm_new = X_logit;
```

```
X_fitglm_new(:,1) = X_fitglm_new(:, 1) + 5;
```

```
p_age_new = predict(mdl_logit, X_fitglm_new);
```

```
age_covariate_effect = mean(p_age_new - p_age)
```

```
X_parents_0 = X_logit;
```

```
X_parents_0(:, 5) = 0;
```

```
X_parents_1 = X_logit;
```

```
X_parents_1(:,5) = 1;
```

```
p_par = predict(mdl_logit, X_parents_0);
```

```
p_par_new = predict(mdl_logit,X_parents_1);
```

```
parents_covariate_effect = mean(p_par_new - p_par)
```

```
%% %Part 5(b)
```

```
clc, clearvars
```

```
%Continuous
```

```
data_ed= readtable("EducationalAttainment.xlsx");
```

```
educLevel= data_ed.educLevel;
```

```
uniqueEducLevels = unique(educLevel);
```

```
educ_counts= histcounts(categorical(educLevel));
```

```
categories(categorical(uniqueEducLevels));
```

```
educ_percent= (educ_counts/sum(educ_counts))*100;
```

```
educLevelstats= [educ_counts, educ_percent]
```

```
famIncome= data_ed.famIncome;
```

```
famIncomestats= [mean(famIncome), std(famIncome)]
```

```
educMother= data_ed.educMother;
```

```
educMotherstats= [mean(educMother), std(educMother)]
```

```
educFather= data_ed.educFather;
```

```
educFatherstats= [mean(educFather),std(educFather)]
```

```
%Categorical
```

```
motherWorked= data_ed.motherWorked;
```

```
tabulate(motherWorked)
```

```
female= data_ed.female;
```

```
tabulate(female)
```

```
black= data_ed.black;
```

```
tabulate(black)
```

```
urban= data_ed.urban;
```

```
tabulate(urban)
```

```
south= data_ed.south;
```

```
tabulate(south)
```

```
age15= data_ed.age15;
```

```
tabulate(age15)
```

```
age16= data_ed.age16;
```

```
tabulate(age16)
```

```
age17= data_ed.age17;
```

```
tabulate(age17)
```

```
%% Part 5(c)
```

```
x_inc= sqrt(famIncome);
```

```
Y= categorical(educLevel);
```

```
X= [x_inc, educMother, educFather, motherWorked, female, black, urban, south, age15, age16, age17];
```

```
mdl_educ = fitmnr(X, Y, 'ModelType', 'ordinal', 'Link','probit');
```

```

fprintf('Ordinal Probit Model Results \n');
results_table = mdl_educ.Coefficients;
disp(results_table);

%hit-rate
predicts= predict(mdl_educ,X);
actual= sum(predicts==Y);
total_predicts= height(Y);
hit_rate= actual/total_predicts;
fprintf('Hit-rate (Accuracy: %.3f)\n', hit_rate);

mcf_r2= mdl_educ.Rsquared.Ordinary;
fprintf('The McFadden R-square comes out to be %f\n', mcf_r2);
%% %Part 5(d)
newinc= famIncome+10000;
x_new_inc= sqrt(newinc);
X_new= X;
X_new(:,1)= x_new_inc;

[~, new_prob]= predict(mdl_educ, X_new);
[~, curr_prob]= predict(mdl_educ, X);

avg_m_effect= mean(new_prob- curr_prob);

fprintf('Avg effect of a $10,000 increase in family income:\n');
cat_names= mdl_educ.ClassNames;
for i=1:length(cat_names)
    fprintf('Change in probability of "%s": %.4f (or %.2f%%)\n',...
        cat_names(i), avg_m_effect(i), avg_m_effect(i)*100);
end

```